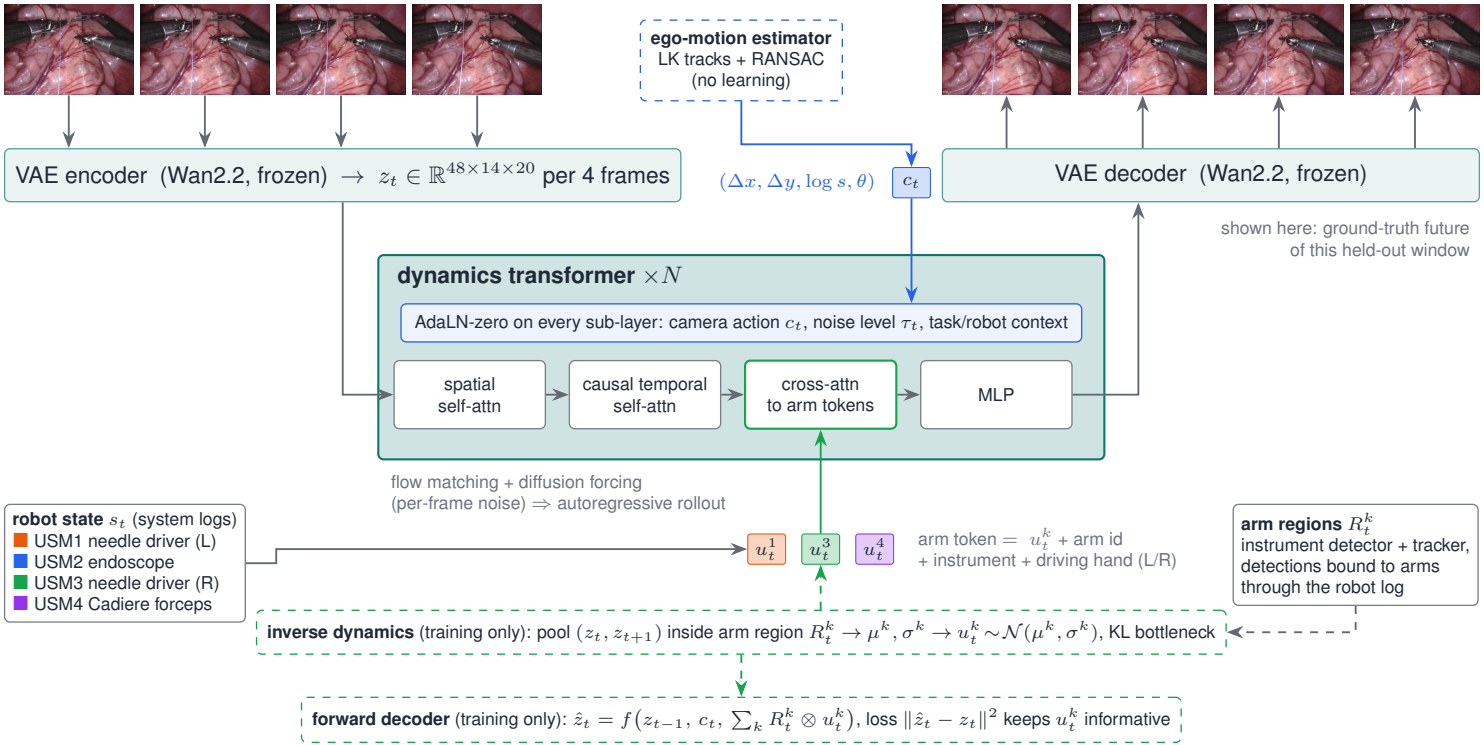

input: context frames $x_{1:4}$ (0.4 s apart) + actions
output: predicted frames $\hat{x}_{5:8}$ (next 1.6 s)
ego-motion estimator
LK tracks + RANSAC
(no learning)
VAE encoder (Wan2.2, frozen) → $z_t \in \mathbb{R}^{48\times14\times20}$ per 4 frames
$(\Delta x, \Delta y, \log s, \theta)$
$c_t$
VAE decoder (Wan2.2, frozen)
shown here: ground-truth future
of this held-out window
dynamics transformer $\times N$
AdaLN-zero on every sub-layer: camera action $c_t$, noise level $\tau_t$, task/robot context
spatial
self-attn
causal temporal
self-attn
cross-attn
to arm tokens
MLP
flow matching + diffusion forcing
(per-frame noise) ⇒ autoregressive rollout
robot state $s_t$ (system logs)
USM1 needle driver (L)
USM2 endoscope
USM3 needle driver (R)
USM4 Cadiere forceps
$u_t^1$
$u_t^3$
$u_t^4$
arm token = $u_t^k$ + arm id
+ instrument + driving hand (L/R)
arm regions $R_t^k$
instrument detector + tracker,
detections bound to arms
through the robot log
inverse dynamics (training only): pool $(z_t, z_{t+1})$ inside arm region $R_t^k \to \mu^k, \sigma^k \to u_t^k \sim \mathcal{N}(\mu^k, \sigma^k)$, KL bottleneck
forward decoder (training only): $\hat{z}_t = f\big(z_{t-1},\, c_t,\, \sum_k R_t^k \otimes u_t^k\big)$, loss $\|\hat{z}_t - z_t\|^2$ keeps $u_t^k$ informative